

University of South Florida

CDA 3201 Logic Design Lab

Lab Report

Experiment	Number	2
	Date	1/25/23
Lab	Section	
	Instructor	Gustavo Olenscki
Report	Due Date	2/1/23
Team	Member #1 Name	Austin Molketin
	Member #2 Name	Diya Jain

Purpose and Objectives

This lab's purpose is to practice logic circuit minimalization and implement a simple SOP circuit. There are 3 problems provided to us by the TA and we are asked to find and draw the minimized circuit (an SOP circuit in all cases) for each problem.

Components Used

We used many components from the mini lab kit provided. The components used are as followed:

- Breadboard
- 74LS04 Inverter gate, 74LS08 Quad AND gate, 74LS32 Quad OR gate
- Many wires
- The power unit
- LED bulb

Design Description

We used a combination of inverters, AND gates, and OR gates to implement the simplification on each problem.

Observations and Data Analysis

Problem #1:

To solve this problem, we first did the Boolean algebra to find the minimized equation and then drew the gate level schematics.

$$f(a,b,c,d) = abc'd' + abcd' + a'bd + abd$$

$$f(a,b,c,d) = abc'd' + abcd' + a'bd + abd$$

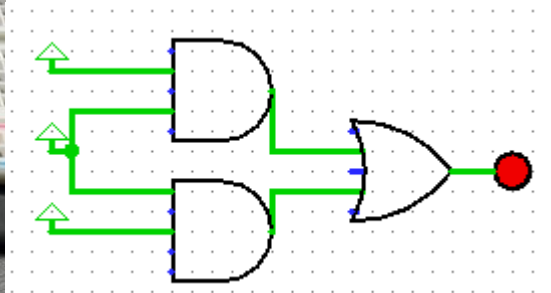
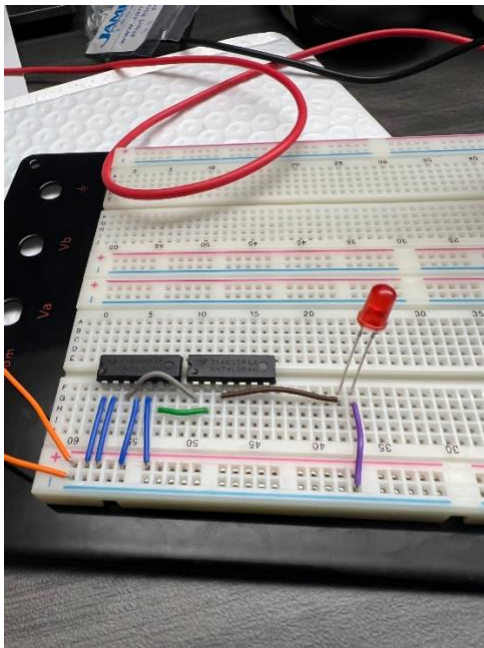
$$f(a,b,c,d) = abd' + a'bd + abd$$

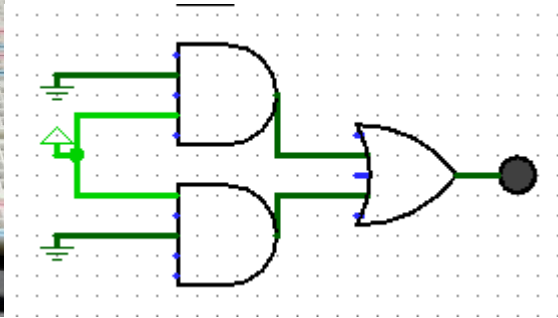
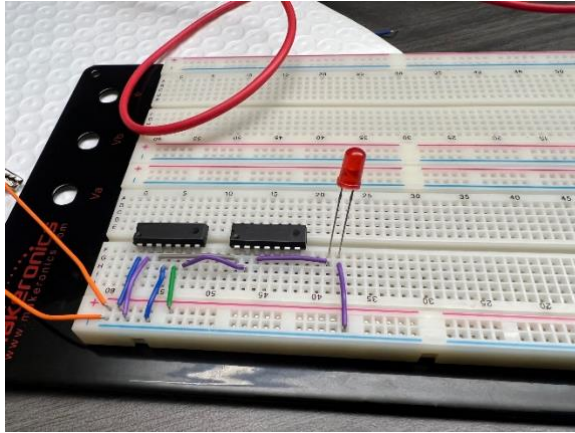
$$f(a,b,c,d) = abd' + abd + a'bd$$

$$f(a,b,c,d) = ab + a'bd$$

$$f(a,b,c,d) = ab + bd$$

***Light was very dim but it was on**





Problem #2:

To solve problem 2, we first made the K-map and found the minimized equation. and then drew the gate level schematics.

a	b	c	Output
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	X ← This is a "don't care" output
1	0	1	X
1	1	0	1
1	1	1	X

bc \ a	0	1
00	1	X (1)
01	1	X (1)
11	0	X (1)
10	0	1

$$f(a,b,c) = a'b'c' + ab'c' + a'b'c + ab'c + abc' + abc$$

$$f(a,b,c) = b'c' + b'c' + a'b'c + abc' + abc$$

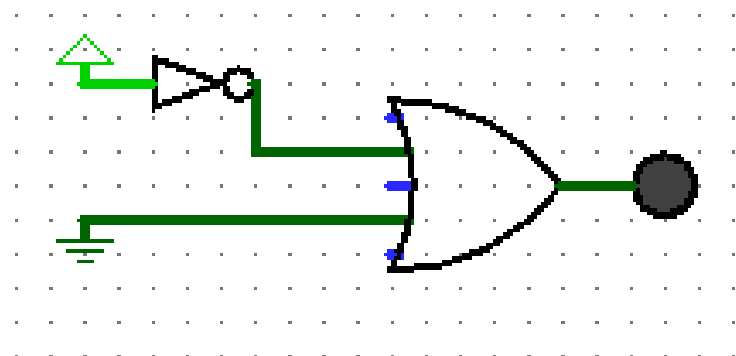
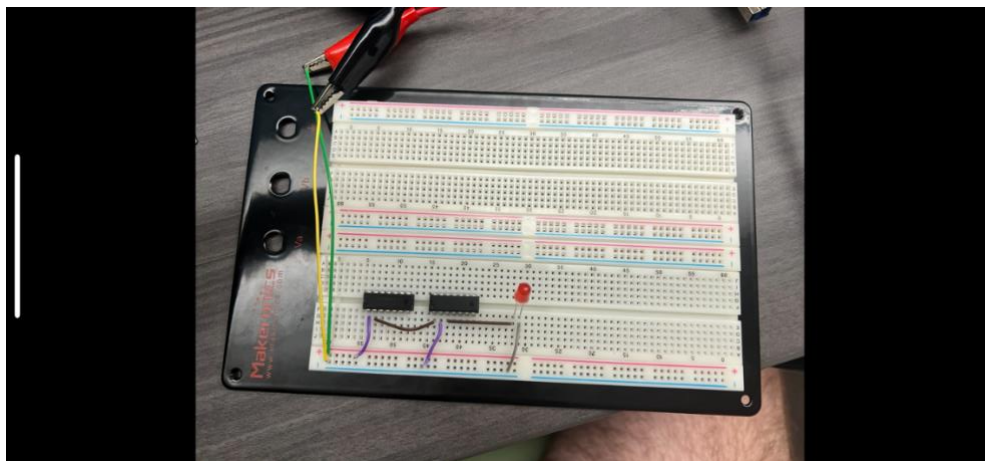
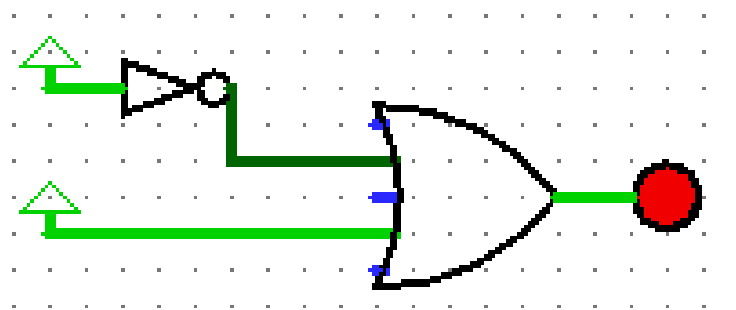
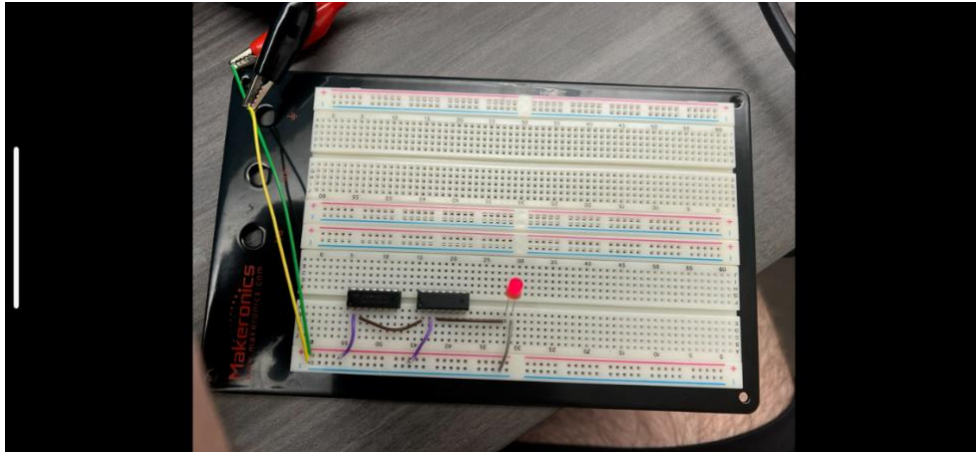
$$f(a,b,c) = b'c' + b'c + b'c + abc' + abc$$

$$f(a,b,c) = b'c' + b'c + abc' + abc$$

$$f(a,b,c) = b' + abc' + abc$$

$$f(a,b,c) = b' + ab$$

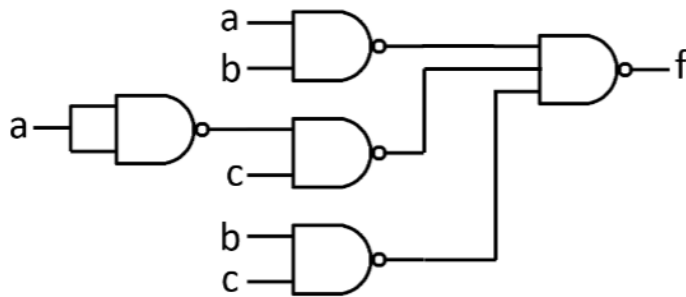
$$f(a,b,c) = b' + a$$



Problem #3:

To solve problem 3, we first analyzed the circuit and marked each step to get to the final value of f and then found the minimized equation.

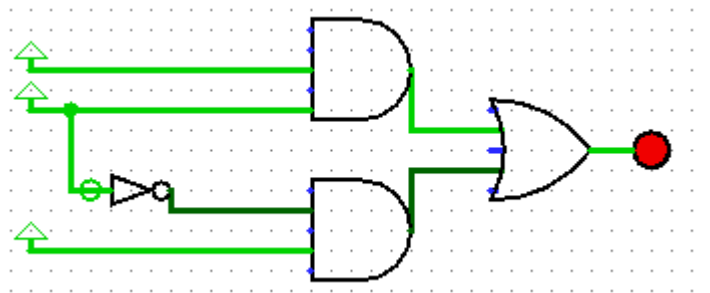
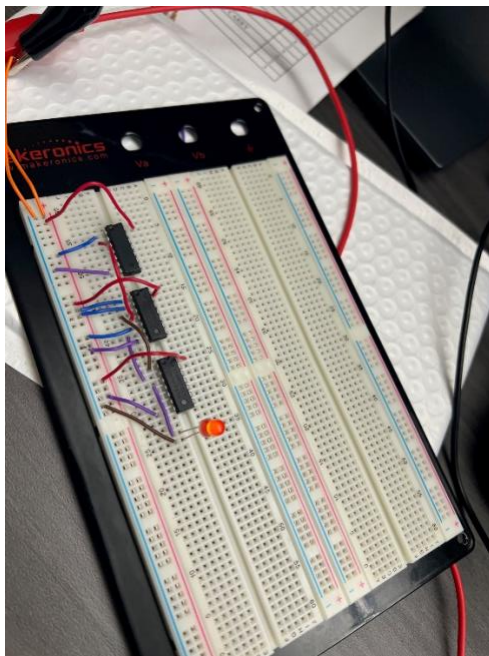
Problem #3:

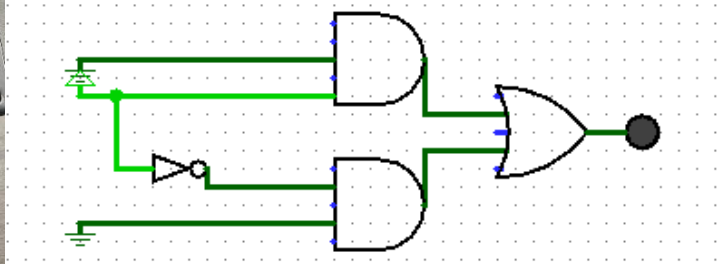
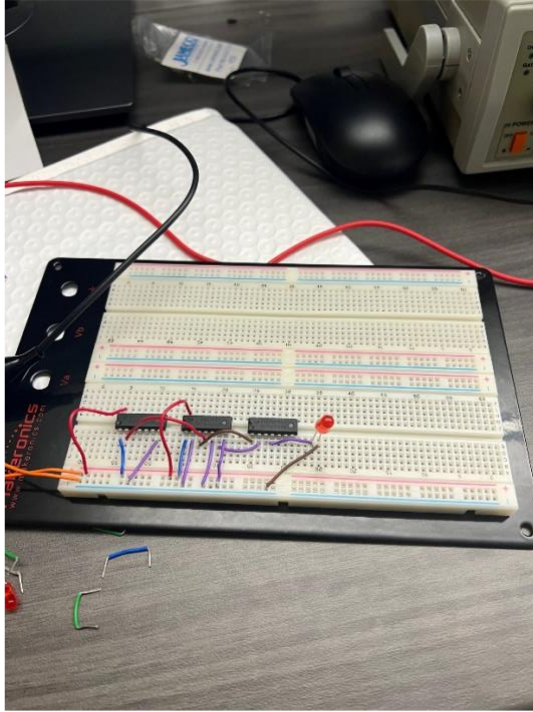


$$f(a,b,c) = ((ab)'((a'c)')(bc)')'$$

$$f(a,b,c) = ab + a'c + bc$$

$$f(a,b,c) = ab + a'c$$





Discussion and Conclusion

In conclusion, this lab's aim was to practice logic circuit minimization and implement a simple SOP circuit using their mini lab kit. We were successfully able to complete all the 3 problems supplied and find their minimized equation. Throughout the experiment we followed all the safety procedures advised by the TA.